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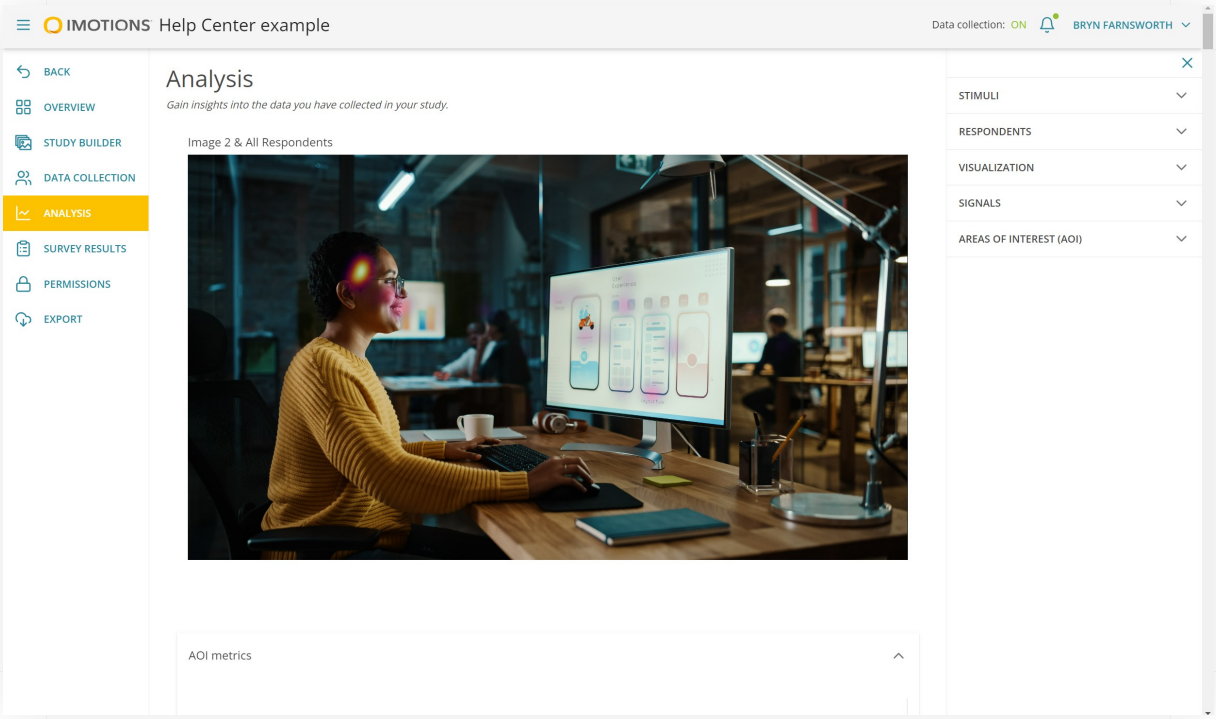
Analysis - iMotions Online 2

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Analysis - iMotions Online

This article explains each of the features associated with the analysis page on iMotions Online.

When you first enter the analysis page, you will be presented with something similar to the following screen:



Center: The center of the screen shows the selected stimulus (in this case an image). As you scroll down, you will see graphs of each collected signal associated with that stimulus. This will be explained in greater detail below, under Signals.

Top: Along the very top of the screen, you can see the study title, which can be changed by clicking on the pencil icon next to it. You can also see whether or not data collection is live (i.e. if a link to study participants is active). You can also access account information, and submit feedback, by clicking on your name in the top right.

Left side: On the left you are able to navigate through other study sections. Each of these are explained in detail within the following articles: Study builder, Data collection, Survey results, and Export.

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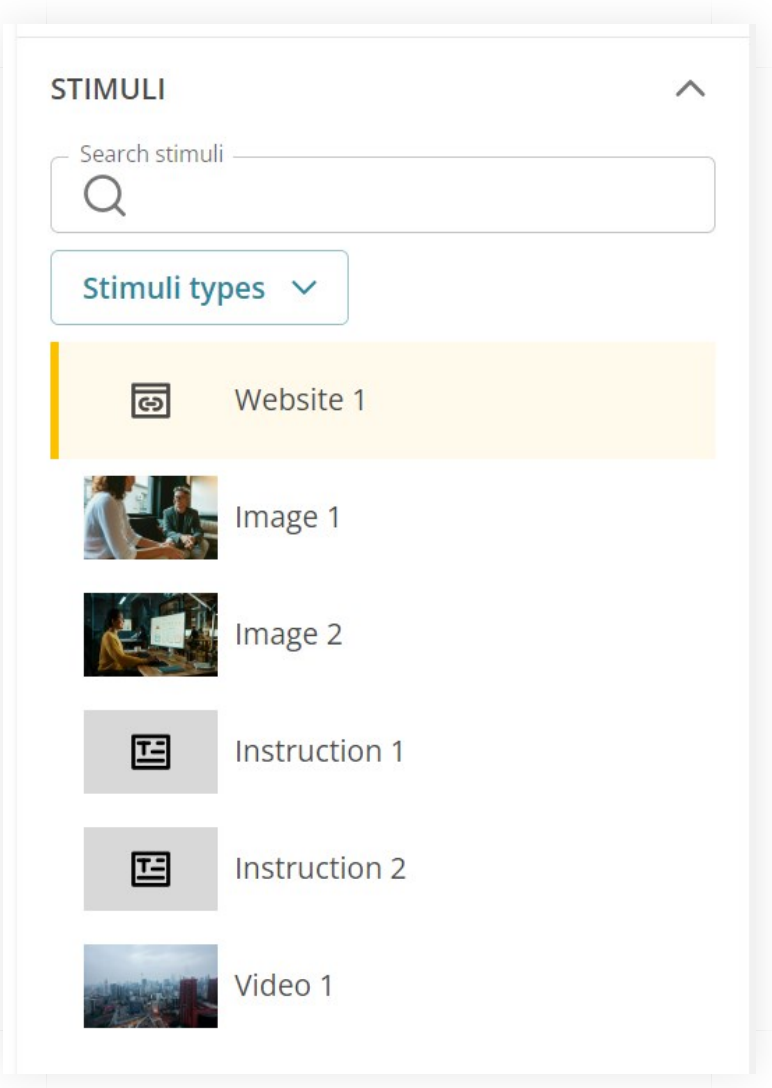
Right side: On the right you can select stimuli, respondents, adjust the visualizations, select specific signals, and create areas of interest (AOIs). More detail for each component is provided directly below, under Settings. You can also collapse and expand this section by clicking on the top right (the cross mark or arrow, respectively).

Settings

Stimuli

Under the stimuli section on the right, it's possible to select which stimulus is brought into focus on the media player in the center of the screen. The text box allows for the stimulus to be searched by name, while the dropdown box ("stimuli types") allows for filtering by stimulus type.

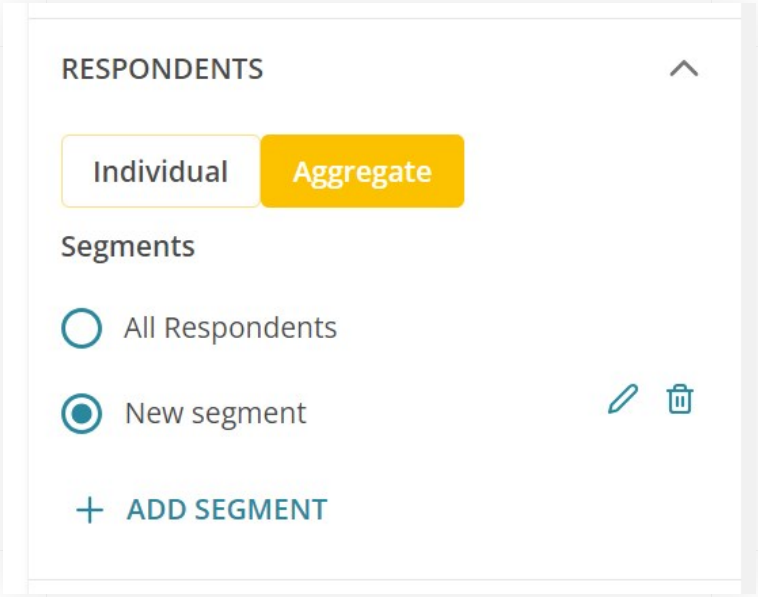
The analysis capabilities will remain the same for each stimulus type, however survey results are shown in the corresponding "survey results" page - you can read about that in more detail within this article.



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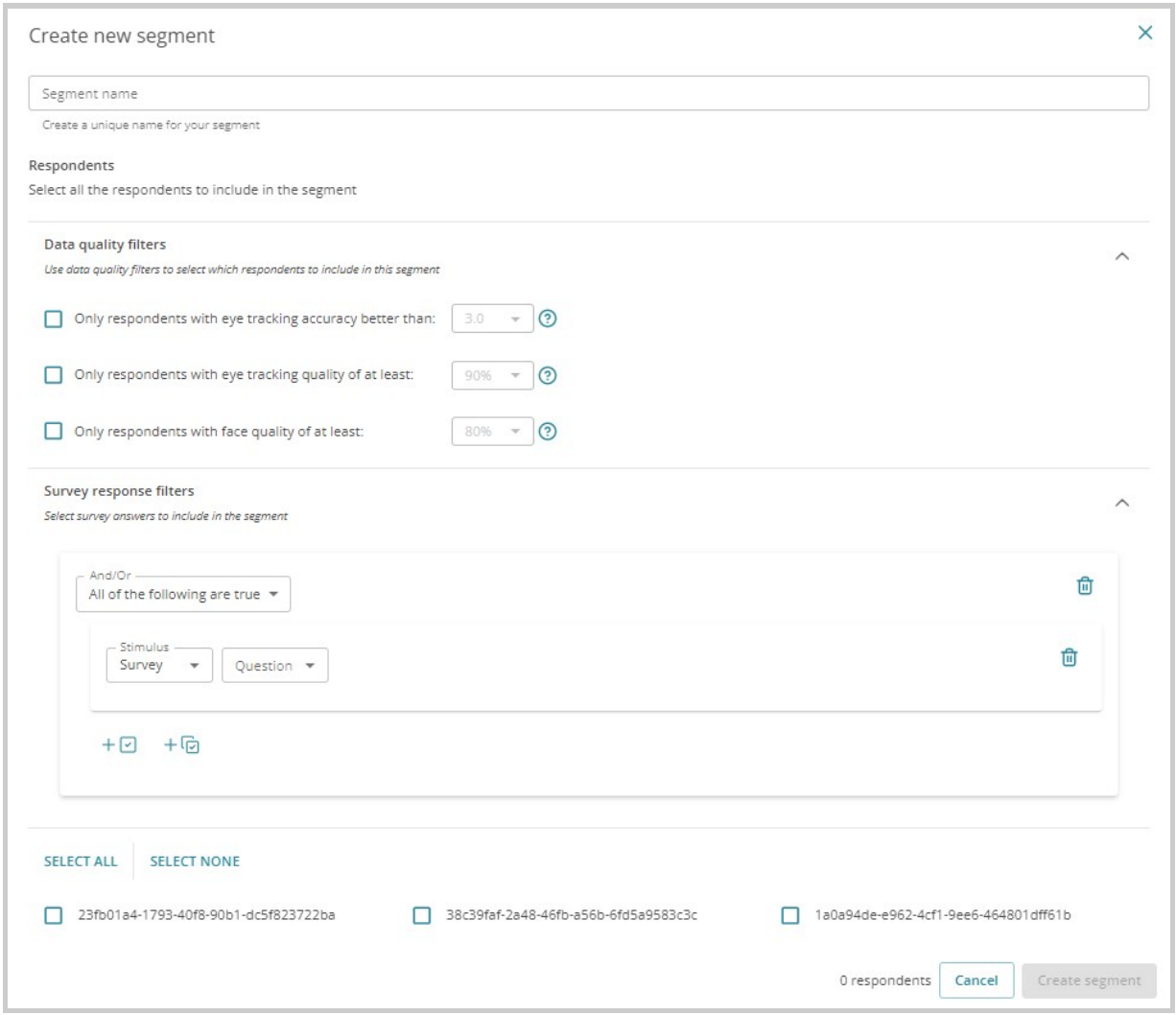
Respondents

The respondents section allows for selection of individual respondents by ID (which can also be seen in the data collection page in more detail). From here it's also possible to aggregate (i.e. combine) the data of multiple participants.



Clicking on "+ add segment" results in the following pop-up:

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This allows you to segment your participants - creating new subgroups from the total participant pool, based on 1.) Data quality filters, 2.) Survey response filters, or 3.) Selection of specific respondents.

A Segment name must be provided in the top text box for any new segments. It's then possible to filter participants based on eye tracking accuracy, eye tracking data quality, and facial expression analysis data quality (and any combination of these elements - each of which are explained in the box below). You can also individually select participants based on their participant ID. Once you have made your segment selections, the relevant participants can be selected from the list (participants who don't match the associated filters will be greyed out) - click "select all" to select those participants. Once you are ready with your participant selections, click on "create segment".

The newly created segment will now appear on the list on the right. If you wish to edit the segment, click on the pencil icon. The segment can also be deleted by click on the trash can icon.

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A note on data quality

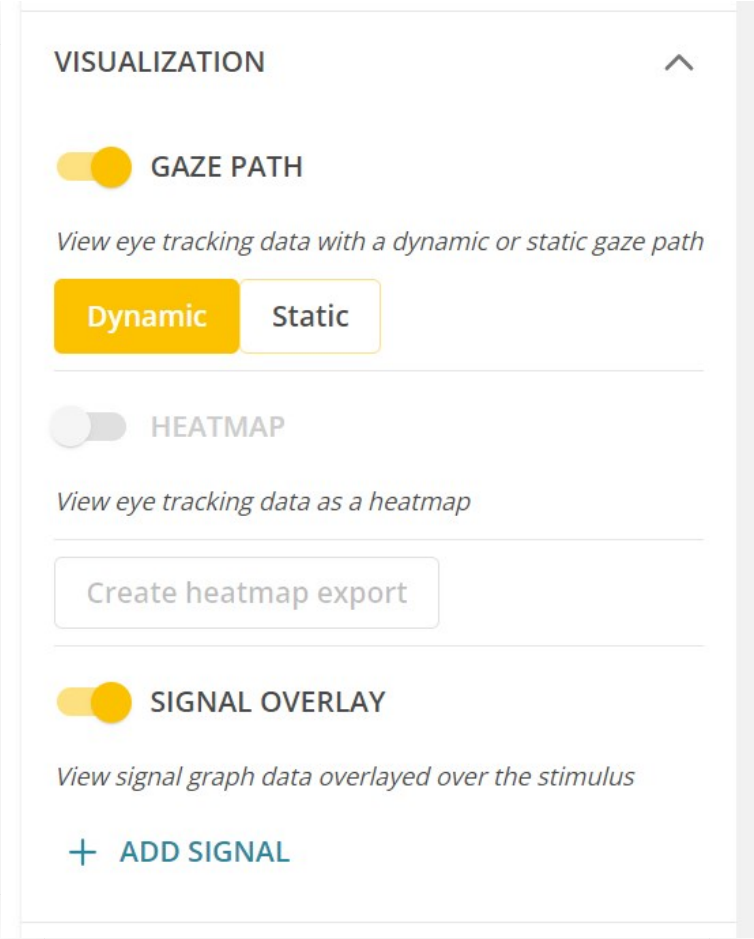
The eye tracking accuracy score is the mean distance between the actual and expected gaze positions calculated during calibration. This distance is measured in visual degrees, with the smaller number meaning more accuracy. Visual degrees is calculated as the angle of the straight lines from the outer edges of the stimulus to the participant's eye (see an example image here). We recommend a threshold of 3.0.
The eye tracking data quality is calculated as the ratio of valid eye tracking samples in relation to the total number of samples collected. For example, if a 1000 samples are collected, but only 800 of them are valid (i.e. where the eyes could be detected as looking at the screen), then the data quality will be 80%. The larger the number, the better the quality. We recommend a quality score of at least 90%.
The facial expression analysis data quality score is calculated as the ratio of valid samples (where the face can be detected) in relation to the total number of samples collected. The larger the number, the better the quality. We recommend a quality score of at least 80%.

Visualizations

The visualizations settings allow for editing of the appearance of the eye tracking and facial expression analysis signals superimposed on the stimulus shown on the media player. These settings will vary depending on whether or not individual or aggregate participant data is selected from the "respondents" settings above. Both versions of the settings (individual and aggregate) are shown below, with each corresponding section (gaze map, heatmap, signal overlay, and create visualization report) explained in turn.

Individual:

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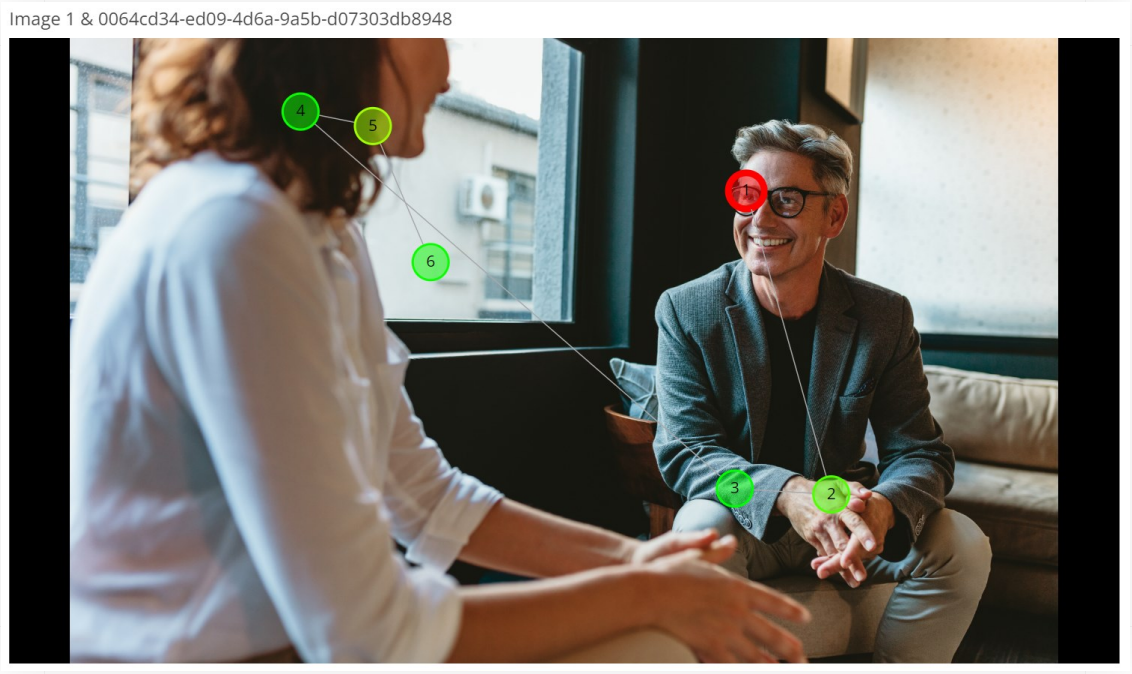
Gaze path:

The "gaze path" refers to visualizing the eye tracking data. This can be turned on or off by pressing the toggle button. When on, this can be viewed as dynamic, in which it's possible to replay the gaze path in a temporal manner (i.e. the gaze can be viewed as though live on the stimulus - note the play button):



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This can also be viewed as static, in which all of the eyes' gaze for the total stimulus exposure time is superimposed upon the stimulus:



The static gaze path shows, by number, the order in which fixations occurred (in the example above, the image begins on the mans eyes, progresses to his hands, and so forth). The color denotes the duration of the fixation, with darker colors representing longer fixation times.

Heatmap:

It is not possible to generate a heatmap for individual participants. The data contained within the static gaze path is essentially the same as a heatmap, but with richer data.

Signal overlay:

The signal overlay provides the possibility to superimpose facial expression analysis, or two types of webcam eye tracking data over the stimulus. Creating a signal overlay will allow you to export the image using the "create visualization report" button, explained in the section below. Click on "+ add signal", and then select the pencil icon - from here you can click on "select" next to "signal" and select a signal from the list. You can also modify the color, and thickness of the line of the signal as it appears over the stimulus here. Each section of selectable signals are explained below:

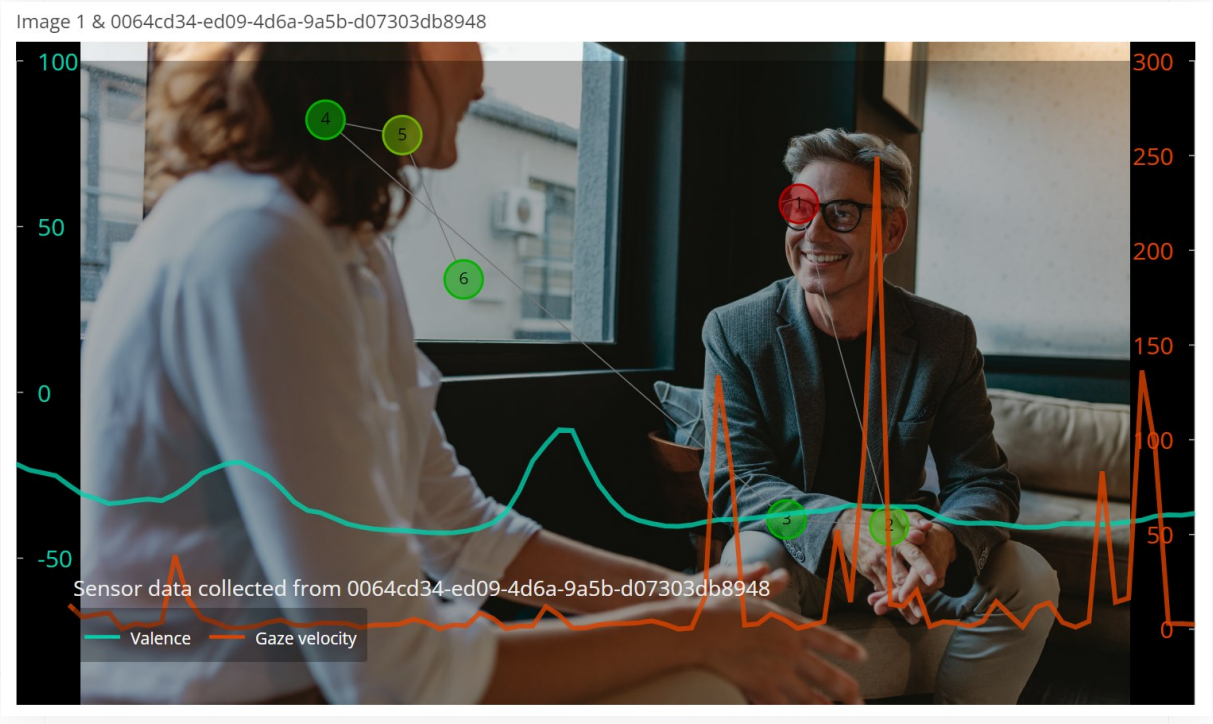
- *Affectiva AFFDEX:* Facial expression analysis is carried out by Affectiva's AFFDEX SDK (version 5.1). You can read more information about this model here. From here you can select from 20 action units (e.g. brow furrow, jaw drop), high level emotional expressions (e.g. joy,

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sadness), as well as head movements (e.g. pitch, yaw), blinks, and overall measurements of valence (positive to negative).

- *I-HMM filter:* The I-HMM filter refers to the Hidden Markov Model for fixation classification. More information about fixation classification with Hidden Markov Models is available here. The signals here are calculated through this method.
- Webcam eye tracker: The signals here relate to data collected by the webcam eye tracking, with no further filtering or processing.

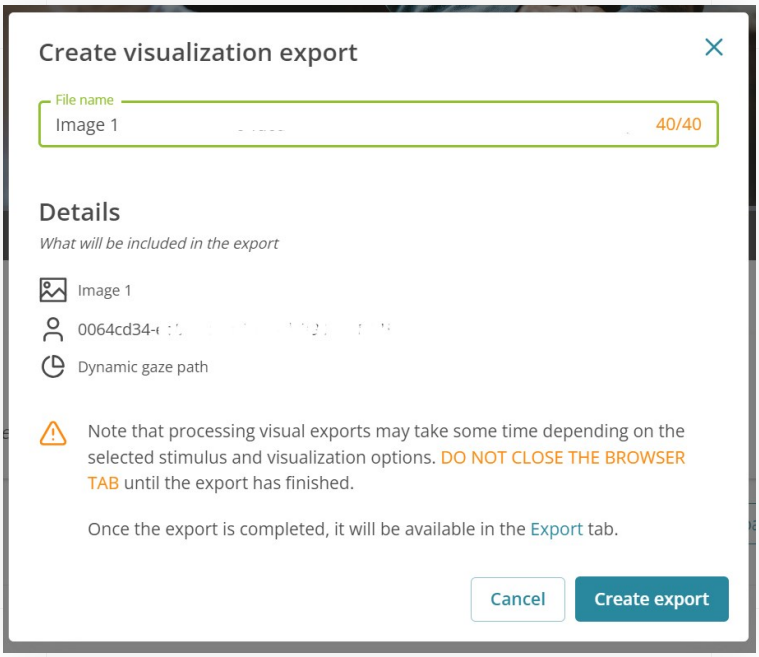
It's possible to select one or two signals to overlay over the stimulus. An example image is shown below in which overall valence from facial expression analysis is shown in turquoise, and gaze velocity is shown in red. Note that the y axes on either side are associated with each signal in the corresponding color. The x axis is the time of the stimulus exposure. The signals are overlaid on the image showing the static gaze path. There is little change in valence throughout the stimulus exposure, which is as expected for a relatively neutral image. The gaze velocity changes in accordance with the fixation duration information obtained from the static gaze path, with faster gaze velocity corresponding to shorter fixations, as the participant looks through different parts of the image.



Create visualization report:

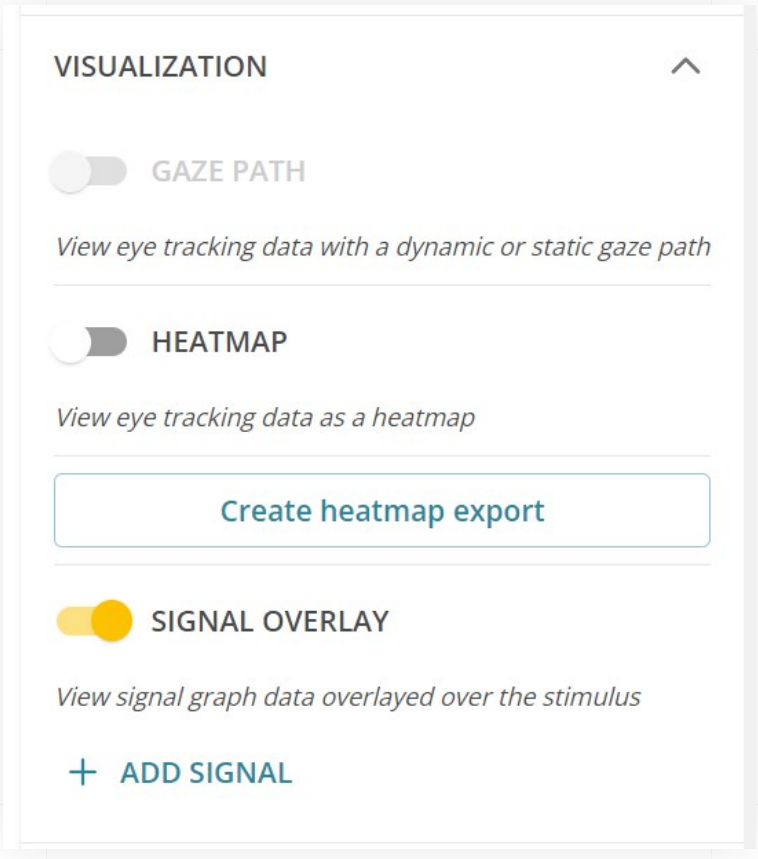
Clicking on "create visualization report" will allow you to export and download the visualization that you can view in the center of the screen. Visualization exports are only available for image or video stimuli, with either the gaze path (for individual participant analyses), heatmap (for aggregate analyses), or the signal overlay enabled. Once you click on the button, you will be presented with the following pop-up box:

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From here you can name the file and create the export. The file will not be downloaded from here, but will instead be sent to the Exports page. You can find more detailed information about the Exports page in this article, although in essence you can simply download the file from there once it has been processed.

Aggregate:



Gaze path:

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It is not possible to generate a gaze path for aggregated data.

Heatmap:

The heatmap shows an aggregated representation of the participants' gazes. As with the gaze path for individual participants, it can be made dynamic, in which it's possible to replay the data in a temporal manner (i.e. the gaze can be viewed as though live on the stimulus - note the play button):



It can also be viewed as static, in which all of the participants' gazes for the total stimulus exposure time is superimposed upon the stimulus:



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The color for the heatmap can also be changed through the dropdown box. The default is set to "Jet", the above images are shown with "iMotions" color scheme.

To download the heatmap images, simply click on "create heatmap export". The image will begin to download automatically.

Signal overlay:

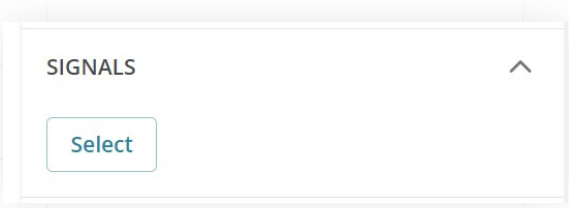
Please see the "signal overlay" section above - the information remains the same for both individual and aggregate data.

Create visualization report:

Please see the "Create visualization report" section above - the information remains the same for both individual and aggregate data.

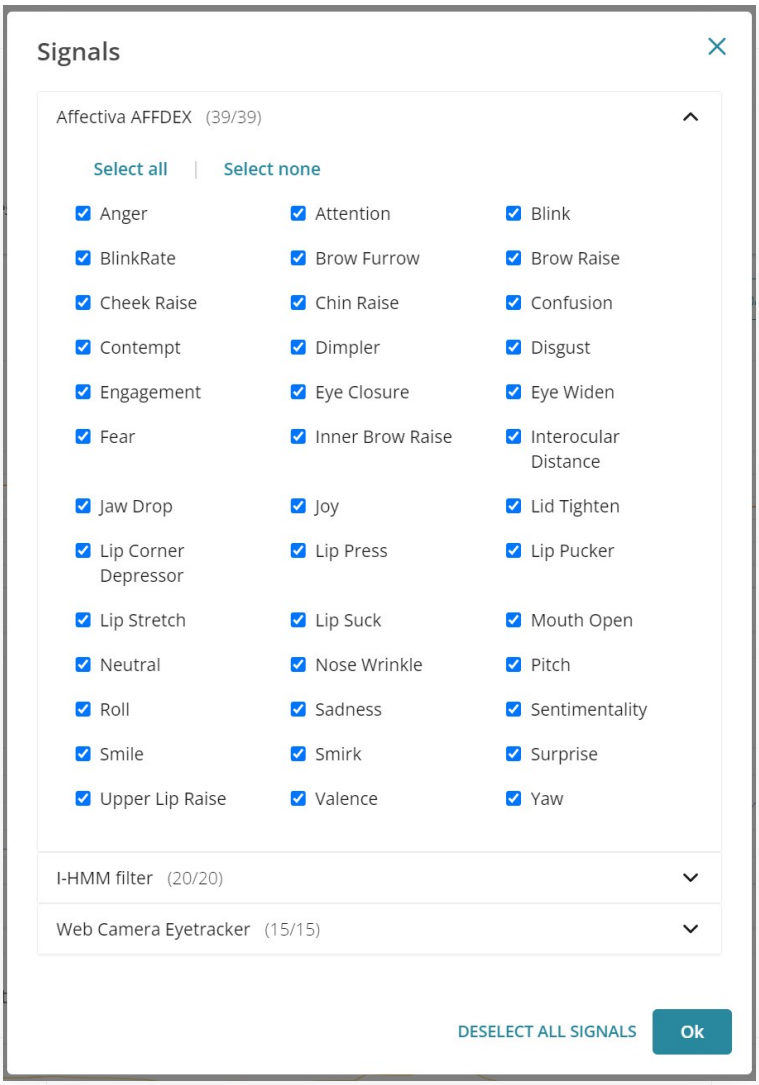
Signals

The signals settings section on the right allows you to select which signals will be shown in the center of the screen, underneath the stimulus.



Press "select" to choose which signals you would like to inspect in detail. A pop-up box will appear. From here it's possible to select individual signals, or all, or none from each section. It's also possible to deselect all signals with the button at the bottom of the pop-up.

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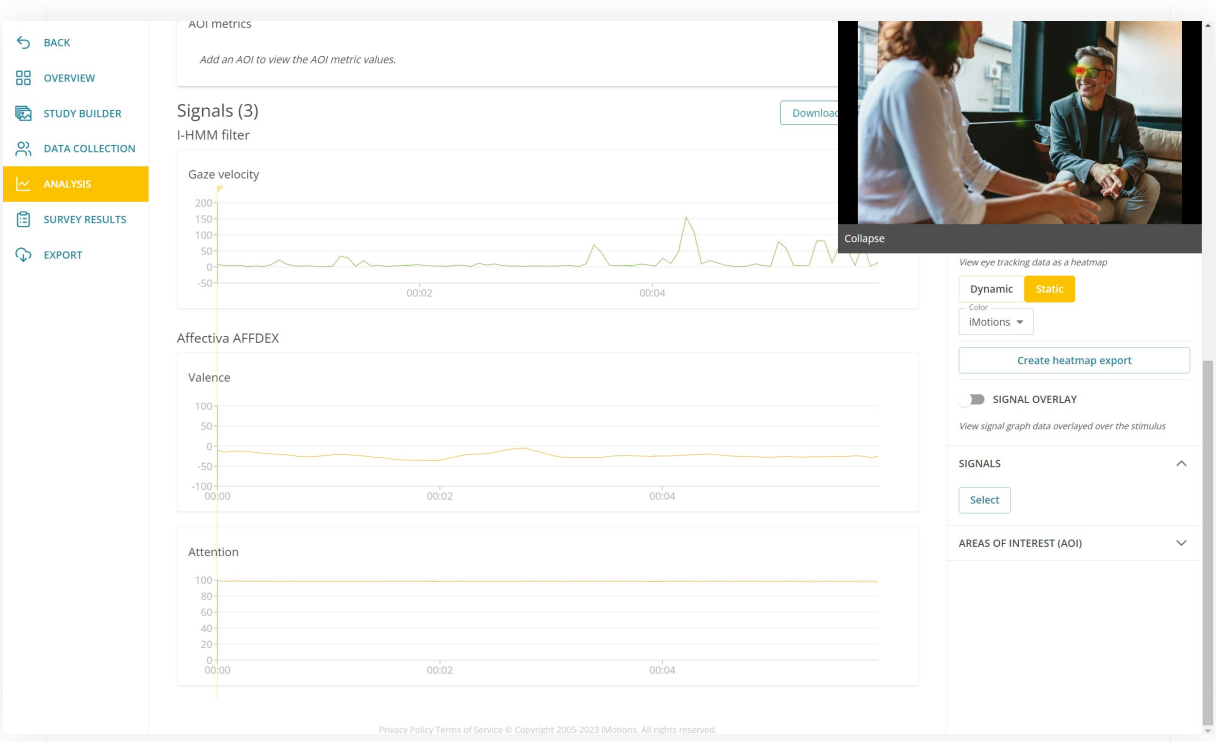


The selections will then appear as line graphs. You can trace the mouse over the lines to view the data values at specific points along the timeline.



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As you scroll down the page, the stimulus will appear in the top right corner. This allows you to view the dynamic changes with the stimulus (if the gaze path or heatmap is set to dynamic) while viewing other signals that are potentially of interest. You can also minimize the stimulus appearance here by clicking on "collapse".



Areas of Interest (AOIs)

Areas of interest allow you to isolate specific visual sections of each stimulus for further detailed analysis. There are two sections under the AOIs settings. The first, AOIs, provides the capability to create (or hide) AOIs. The second, AOI metrics, allows you to calculate metrics and download the resulting data. Each part is explained below:

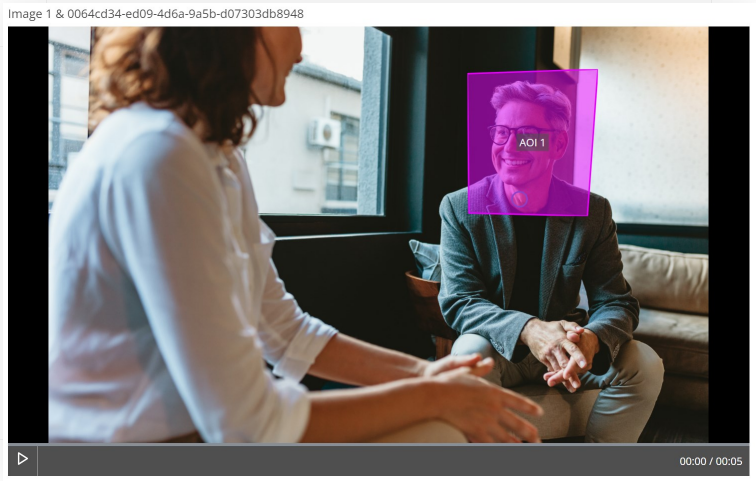
AOIs

To create an AOI, click "+ add AOI". Now place the mouse over the stimulus in the center of the page - from here you can trace around the area of interest. The cursor will have become a cross, allowing you to place points. Each point that is placed will provide an attached line - clicking again will anchor that line in position, providing another attached line, and so on. Once you have surrounded the area of interest, you can join the lines together to form the desired shape by returning the final line to the starting point. The final point will glow slightly as it is connected to the starting point. Please note that there may be a slight delay before the AOI appears once the joining of the final point has taken place. The sequence is summarized in the images below:

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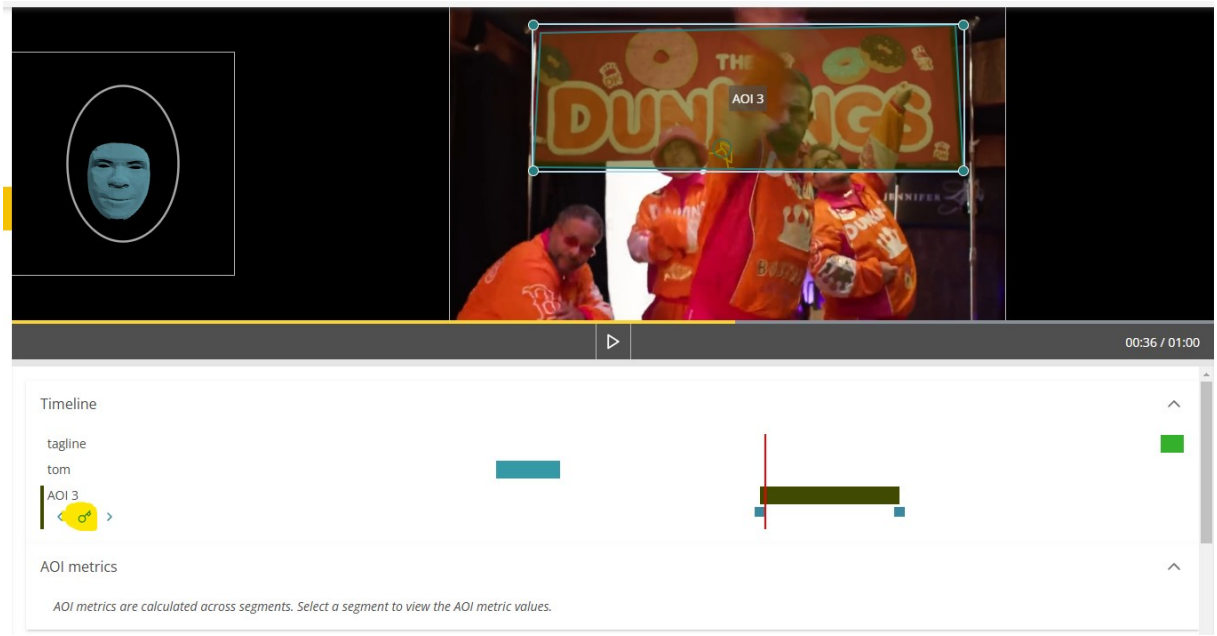


Multiple AOIs can be created on the same stimulus. The name, and color surrounding each can be changed by clicking on the pencil icon to the right of each AOI. These can also be individually hidden or displayed by clicking on the eye icon to the left of the AOI name, or deleted by clicking on the trash can icon. All of the AOIs can also be shown or hidden by clicking on "show all" or "hide all" respectively.

Video AOIs

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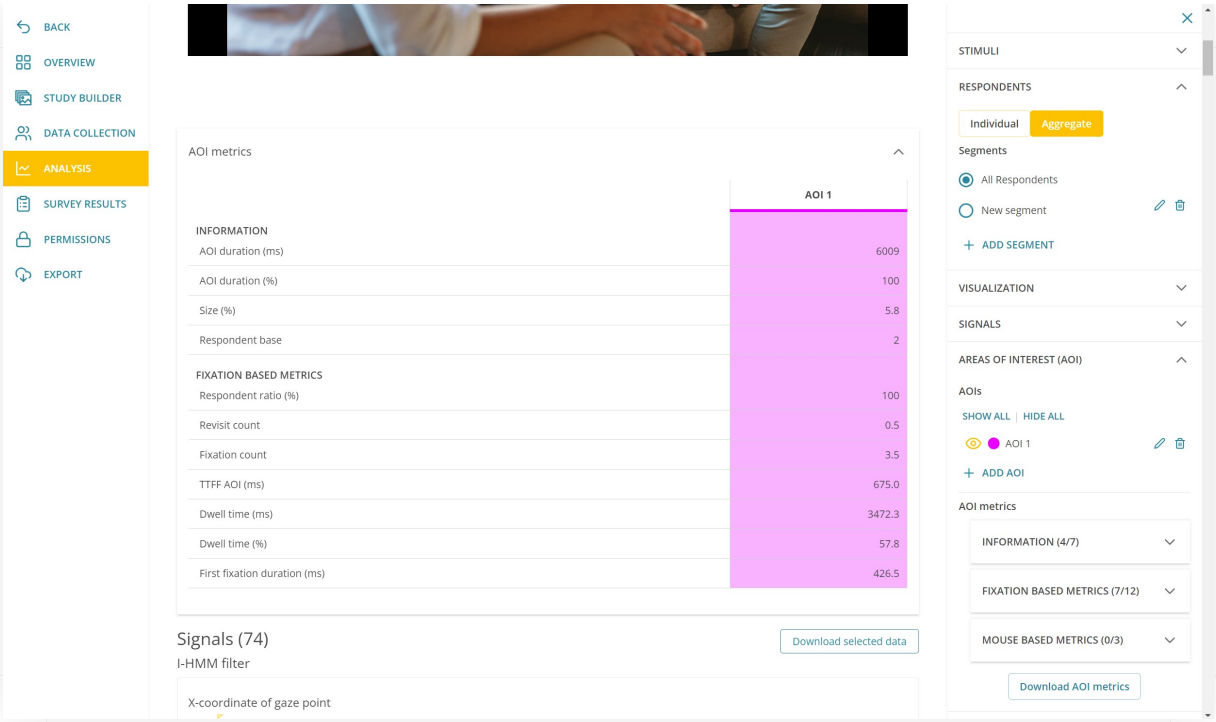
AOIs on videos can be added in the same way as static images. After navigating to the part of the video your target appears you can add your AOI from the panel as usual. Under the replay, you can open the timeline to see the control area for your AOI. As your target moves or changes size, you can add additional keyframes by adjusting your AOI and then pressing the Key button to add it as a keyframe. This will add a blue square marker to the lane below the AOI timeline. When the target is no longer on the screen you can add a keyframe without adjusting the AOI which will trigger the AOI to turn off.



AOI metrics

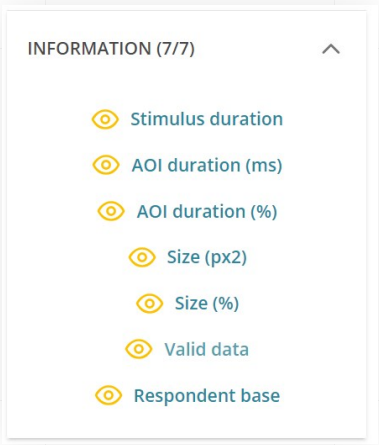
After an AOI is added, AOI metrics will be calculated and will appear in the center of the screen under the Aggregate view from the Respondents Panel, below the stimulus, and above any selected signals. This may take several minutes to calculate depending on the amount of data and your internet speed.

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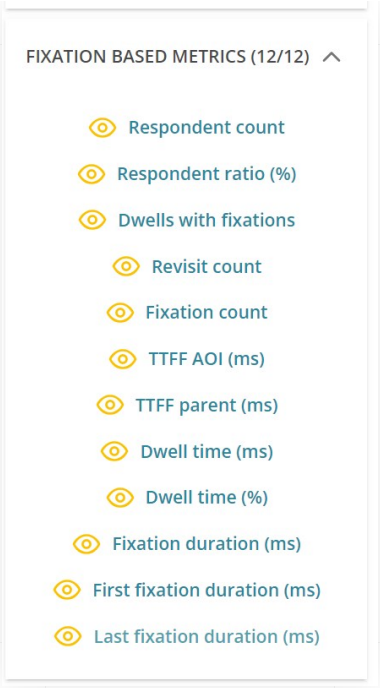
The option to select specific AOI metrics will also appear within three dropdown boxes underneath "AOI metrics". These are:

- Information - allowing the selection of information related to the AOI itself:

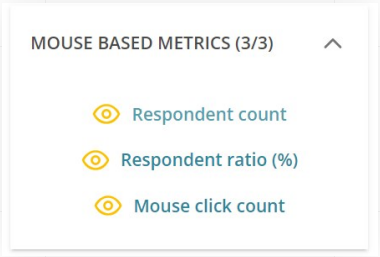


- Fixation-based metrics

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- Mouse-based metrics



More information about AOI metrics is available here. Note that not all metrics are applicable to webcam-based eye tracking, and are therefore not in the list of AOI metrics.

Each selected metric will appear in the list of graphs in the center of the screen.

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